Auditing Closed-Loop Learning in Recurrent Neural Networks: Reproduction, Robustness, and Generalization

Anonymous authors
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Abstract

Recurrent neural networks are often used as mechanistic models of learning and control, but closed-loop training creates reproducibility challenges because a model's actions alter future inputs. We conduct a claim-level reproducibility study of Ger and Barak's closed-loop RNN learning dynamics, testing independent implementation, seed variation, protocol perturbations, coupled-system diagnostics, and architecture/task transfer. The central deployed-loss gap appears in our original-setting implementation: 50/50 paired seeds favor closed-loop training, with mean final deployed losses 0.0933 versus 0.3119. However, two original signatures are only partially recovered: our loss-only stage detector is not a test of the paper's spectral stage definition, and our open-loop curves do not reproduce the upstream artifact's post-initial sharp peak. A targeted A1 analysis separates stability and behavioral tradeoffs: short-horizon improvements coincide with coupled-radius increases in 120/120 runs, while the loss-only short-vs-long criterion is supported in 0/120. Generalization is hierarchical: architecture transfer within the double-integrator-style task is strong, tracking transfer is partial, and path-integration transfer is weak. These results motivate reporting practices for closed-loop RNN studies: deployed closed-loop loss, peak signatures, paired seeds, spectral stage criteria, coupled-system spectra, feedback strength, rollout horizon, and failure rates.

1 Introduction

Closed-loop recurrent neural networks occupy an important middle ground between supervised sequence modeling, control, and computational neuroscience. In a closed-loop environment, an RNN's output changes the future inputs it receives. This feedback makes the training problem qualitatively different from open-loop imitation or teacher-forced supervised learning, where future inputs are independent of the learner's deployed actions. Ger and Barak (Ger and Barak, 2025) recently argued that this distinction produces learning dynamics that are fundamentally different from those of otherwise identical open-loop RNNs. Their paper is valuable because it makes a mechanistic claim rather than only a performance claim: the coupled agent-environment system, not the RNN alone, determines important learning transitions.

This paper is not a figure-level reproduction study. Instead, we use Ger and Barak's closed-loop RNN learning framework as a test case for claim-level reproducibility in recurrent control systems. We ask which conclusions survive independent implementation, seed variation, optimizer and hyperparameter perturbations, coupled-system stability analysis, and transfer to nonlinear or path-integration-style settings. This framing matters for TMLR-style reproducibility: a rerun of a public artifact can be educational, but it is only broadly useful when it surfaces reusable lessons about when claims are reliable, when they are protocol-sensitive, and what future papers should report.

Our study makes four contributions. First, we provide an independent, config-driven reproduction of the central open-loop/closed-loop learning comparisons in Ger and Barak. Second, we convert qualitative claims about learning stages and stability transitions into quantitative, seed-level tests with confidence intervals. Third, we measure robustness under perturbations to optimizer, initialization, feedback strength, noise, and rollout horizon, identifying which conclusions are stable and which are protocol-sensitive. Fourth, we distill an audit checklist for future closed-loop RNN studies, including paired-seed comparisons, deployed closed-loop loss, stage criteria, coupled-system stability diagnostics, and failure-rate reporting.

2 Related Work

Task-trained RNNs in computational neuroscience. Task-trained RNNs are widely used as mechanistic models because they can solve cognitive and motor tasks while exposing internal dynamics for analysis. Early and influential work showed that high-dimensional trained RNNs can implement low-dimensional dynamical mechanisms (Sussillo and Barak, 2013), and that recurrent dynamics can account for context-dependent computation in prefrontal cortex (Mante et al., 2013). This line of work has become a general modeling strategy in systems neuroscience, where RNNs serve as hypothesis-generating models rather than only predictive tools (Barak, 2017). At the same time, large-scale analyses of trained recurrent networks show that architecture and training details can change representational geometry and dynamics even when task performance is similar (Maheswaranathan et al., 2019). This motivates our claim-level framing: reproducing a task curve is not enough if the mechanistic conclusion depends on training protocol or model class.

Dynamical-systems analysis of RNNs. The original paper sits in a tradition of reverse-engineering trained RNNs through fixed points, low-dimensional projections, eigenvalues, and linearized dynamics. Sussillo and Barak's low-dimensional analysis made the case that trained recurrent networks can be understood through dynamical-systems tools (Sussillo and Barak, 2013), and later work used related analyses to compare recurrent dynamics across large populations of networks (Maheswaranathan et al., 2019). Our audit differs in emphasis. We do not only ask whether a trained RNN has interpretable internal dynamics; we ask whether the relevant dynamical object is the RNN alone or the coupled agent-environment system. This distinction is central in closed-loop settings because the environment transforms actions into future observations.

Open-loop training, closed-loop deployment, and distribution shift. The mismatch between teacher-forced training and deployed rollouts is well known in sequence prediction and imitation learning. Scheduled sampling was proposed to reduce the discrepancy between training on ground-truth previous tokens and testing on model-generated tokens (Bengio et al., 2015). In imitation learning, DAgger and related reductions explicitly address the fact that a learner's actions change the future states it visits, invalidating i.i.d. supervised-learning assumptions (Ross et al., 2011). Closed-loop RNN learning has an analogous but mechanistically richer mismatch: open-loop imitation can minimize teacher-forced action error while still failing under deployed closed-loop rollouts. This connection motivates our insistence on reporting deployed closed-loop loss, not only open-loop or imitation loss.

Reproducibility and robustness in machine learning. Our audit follows the broader shift from artifact reruns to structured reproducibility studies. The NeurIPS reproducibility program emphasized clear artifacts, documented environments, and reproducibility checklists (Pineau et al., 2021). In deep reinforcement learning, Henderson et al. showed that algorithmic conclusions can be sensitive to seeds, evaluation protocol, and implementation details (Henderson et al., 2018). Closed-loop RNN studies share many of these risks because training outcomes can depend on random initialization, rollout horizon, optimizer, and environment coupling. The present paper therefore treats Ger and Barak's work as a case study for a reusable audit protocol: paired seeds, stress tests, coupled-system diagnostics, and reporting lessons.

We use *reproduction* for tests that preserve the original task family and claim, *robustness* for perturbations of seeds, optimizer, initialization, horizon, noise, feedback, and clipping, and *generalization* for architecture or task-family changes. This terminology is intentionally conservative: a robustness or generalization failure does not by itself falsify the original result, but it does delimit where the claim should be reported as reliable.

3 Original Claims and Audit Questions

We separate the original claims under reproduction from audit questions introduced by this study. The reproduced claims are C1–C3. C1 is the central divergence claim: identical RNNs trained open-loop and

Table 1: Direct mapping from original evidence to our reproduction tests. This separates figure-level reproduction from claim-level decisions.

Original claim	**Original evidence**	**Our test**	**Result**	**Decision**
Closed/open divergence	Loss/gain figures and open-loop sharp peak	50 paired seeds plus peak-signature check	final gap 50/50; post-initial peak 0/50	Partial
Stage structure	Spectral stages: negative-position policy, stability phase, refinement	Loss-only derivative and change-point observer	slow phase 50/50; spectral C2 untested	Loss-only audit
Coupled stability	Coupled eigenvalue argument and Stage 2 alignment	Coupled radius crossing vs loss changepoints	50/50 crossings; median 201 epochs before τ_2	Partial
Motor-control applicability	Tracking extension	Tracking plus ring variants	tracking 8/10; hard ring 4/40	Partial boundary

closed-loop follow different learning trajectories and differ under closed-loop deployment. C2 is the stage claim: closed-loop learning passes through distinct learning stages. C3 is the mechanistic stability claim: progress depends on stability of the coupled agent-environment system.

The audit questions are A1–A2. A1 asks whether C1–C3 are robust to protocol perturbations and whether a targeted short-vs-long horizon analysis reveals the expected stability tradeoff. A2 asks whether the phenomena generalize across architectures and tasks. We use A1 and A2 to avoid implying that Ger and Barak made every robustness or generalization claim in exactly our framing.

For each reproduced claim or audit question, we define three evidence levels. A result is *reproduced* when the effect and diagnostic hold under the relevant protocol for most paired seeds, with confidence intervals excluding zero or a pre-specified smallest effect size of interest. For C1 this smallest effect is a 5 percent relative deployed-loss gap, corresponding to roughly 0.0047 absolute loss in the original-setting closed-loop scale; smaller gaps are reported continuously but not counted as deployed-loss divergence. A result is *partially reproduced* when the direction appears but is unstable across seeds, sensitive to plausible perturbations, or only weakly supported by the diagnostic. A result is *not reproduced* when the effect is absent, reverses, or depends on hand-picked visual criteria.

4 Closed-Loop RNN Reproducibility Audit Protocol

We specify a modest audit protocol rather than proposing a new benchmark. The protocol is a recipe for testing claims about closed-loop RNN learning dynamics in any recurrent control setting where actions can affect future observations. Its inputs are: a set of original claims, a task/environment, an open-loop training procedure, a closed-loop training procedure, a seed set, a small set of protocol perturbations, and at least one held-out architecture or task variant. Its outputs are: per-seed deployed closed-loop metrics, stage labels, stability diagnostics, perturbation summaries, generalization summaries, and a claim-level reproducibility matrix. Table 2 gives the concrete specification used in this paper.

The protocol deliberately separates *measurement* from *interpretation*. Measurement consists of paired runs, deployed losses, spectra, stage labels, and perturbation outcomes. Interpretation happens only after aggregating seed-level outputs into claim-level decisions. This separation is important because closed-loop RNN papers often make qualitative statements about stages or mechanisms from a small number of curves; our protocol requires those statements to be tied to explicit detectors, uncertainty estimates, and failure rates.

4.1 Metrics and run accounting

The main metrics are deployed closed-loop loss, teacher-forced imitation loss, effective feedback gain, coupled and RNN-only spectral radius, algorithmic stage labels, and A1 multi-horizon tradeoff components. Exact formulas are in Appendix A. The 455 completed paired runs consist of 50 core C1–C3 runs, 195 robustness runs, 120 targeted A1 multi-horizon runs, and 90 A2 extension runs. Detector thresholds were selected during smoke-test development before running the full reported sweeps; threshold sensitivity is reported to show which conclusions depend on those choices.

Table 2: Closed-loop RNN reproducibility audit protocol. The protocol is intentionally lightweight: each row defines a required audit step, its minimum implementation, and the output needed to support claim-level conclusions.

Audit step	Minimum requirement	Recorded output
Paired training	Clone the same initialization into open-loop and closed-loop learners; keep architecture and evaluation budget fixed.	Per-seed train loss, deployed closed-loop test loss, peak-over-training deployed loss, final loss, and effective-gain trajectory for both learners.
Seed-level uncertainty	Run enough seeds to estimate direction agreement and confidence intervals; report seed count explicitly.	Mean, 95 percent bootstrap CI, fraction of seeds supporting each claim, and seed-level failure/recovery rates.
Stage detection	Predefine detectors in the same diagnostic space as the claim; for this audit, separate loss-only stages from spectral stages.	Stage labels, plateau duration, transition epochs, spectral-crossing gaps, and frequency of detected stages across seeds/settings.
Coupled diagnostics	Compute RNN-only and coupled agent-environment spectra whenever the system can be linearized or locally linearized.	Spectral radius time series, stability-crossing epoch, plateau-exit gap, and predictive correlation with recovery/failure.
Robustness perturbations	Change factors that alter optimization or environment coupling, such as optimizer, initialization scale, feedback strength, noise, horizon, control penalty, and clipping.	Claim support under each perturbation, effect-size changes, and perturbation-level failure rates.
Generalization check	Test at least one architecture or task variant that preserves action-dependent inputs and one that changes the recurrent/control structure.	Claim support by variant and a boundary-condition statement explaining when the phenomenon does or does not transfer.
Claim/audit decision	Classify each claim or audit question as reproduced, partially reproduced, or not reproduced using the same criteria across settings.	Matrix linking evidence type, original setting, robustness, generalization, and actionable lesson.

4.2 Paired open-loop and closed-loop training

For each seed, one initialization is cloned into an open-loop and a closed-loop learner. The closed-loop model is trained by differentiating through the rollout with backpropagation through time, matching the direct-gradient setup of the public artifact; this is not REINFORCE-style stochastic policy-gradient estimation. The open-loop model imitates teacher trajectories in our independent implementation. Both are evaluated under deployed closed-loop rollout.

4.3 Stage detection

The original paper anchors its stages spectrally: Stage 1 is a negative-position policy with an unstable complex pair, Stage 2 ends when the dominant eigenvalues of the coupled matrix enter the unit disk, and Stage 3 involves policy refinement and growth of a third real mode. Our saved independent runs do not store the full eigenvalue trajectories needed to test that exact spectral definition. We therefore report a narrower loss-only observer: whether derivatives and changepoints of deployed loss recover corresponding stage boundaries. A failure of this detector is not a falsification of the original spectral C2 claim.

4.4 Coupled-system stability diagnostics

For linearized settings, we compute the spectral radius of the RNN recurrent matrix and of the coupled agent-environment transition matrix. This targets C3 directly: if closed-loop dynamics are governed by the agent-environment loop, the coupled diagnostic should be more informative than an isolated RNN diagnostic.

4.5 Robustness and generalization

The A1 robustness sweep changes learning rate, initialization scale, feedback strength, episode length, observation noise, control penalty, optimizer, and gradient clipping. The Adam condition is included because Ger and Barak's Appendix C.7 predicts that adaptive optimization mitigates the structured short/long-horizon conflict. The A2 generalization sweep tests nonlinear RNNs, GRUs, low-rank RNNs, tracking control, and ring/path-integration-style tasks.

4.6 Original artifact, independent implementation, and deviations

We preserved the public artifact under `external/original_artifact/`. It contains the original notebooks for figure reproduction, nonlinear training, theory, and tracking, plus the pickle files used by the artifact. Our reproduction log records which files load, which notebooks depend on precomputed data, and which notebook runs are long or checkpoint-heavy.

The experiments reported here use an independent, config-driven implementation rather than direct notebook execution. The double-integrator task, paired open/closed training, effective gains, coupled spectra, stage detectors, robustness perturbations, and generalization tasks were reimplemented as package modules and command-line scripts. We matched the original double-integrator architecture and training profile where needed for C1–C3, then intentionally introduced new A1/A2 audits over seeds, optimizer, initialization, feedback, horizon, noise, clipping, architecture, and task. We did not contact the original authors; all choices were made from the paper, public artifact, and documented configs. Exact notebook reruns are therefore treated as artifact checks, while the paper's quantitative claims are evaluated through the independent implementation. Every table and figure can be regenerated from saved raw CSVs with `scripts/run_full_audit.sh`, `scripts/run_targeted_c2_a1_a2.sh`, and `python -m closed_loop_repro.plotting.make_all_figures -config configs/figures/tmlr.yaml`.

A code-level comparison shows that the core reproduction matches the original double-integrator dynamics, hidden size, tanh controller class, SGD learning rate, batch size, rollout horizon, gradient clipping, and teacher-student design. It is not a line-for-line reimplementation: the original nonlinear notebook uses a zero first action, current-state loss accumulation, white-noise teacher inputs for open-loop training, 1001 epochs, and smaller input/output initializations under its small-scale setting. The paper states an unregularized objective, while the public nonlinear notebook and our main config both use a small control penalty. We therefore interpret the package as an independent claim-level audit of the original protocol family, not an exact numerical rerun of every notebook.

5 Result 1: Direct Reproduction of the Core Divergence

The final deployed open-loop/closed-loop gap appears cleanly under our original-setting double-integrator implementation (Figure 1). Across 50 paired seeds, the closed-loop trained model reaches final deployed loss 0.0933 with bootstrap 95 percent CI $[0.0932, 0.0935]$, seed SD 0.00054, and IQR $[0.0929, 0.0937]$. The open-loop trained student, evaluated under deployed rollout, has final loss 0.3119 with CI $[0.3044, 0.3193]$, seed SD 0.0269, and IQR $[0.2919, 0.3255]$. The paired gap is 0.2186 with CI $[0.2110, 0.2260]$ and direction agreement in 50/50 seeds. Effective feedback gains also diverge: the final closed/open gain distance is 0.5045 with CI $[0.4998, 0.5092]$.

This supports the broad C1 warning that teacher-forced performance can mispredict deployed behavior, but it does not reproduce every visual signature of the original figure. In the upstream artifact, the open-loop deployed-loss curve has a post-initial sharp peak at epoch 221, with peak/initial ratio 5.24. In our independent original-setting runs, the maximum open-loop deployed loss occurs at epoch 0 in 50/50 seeds; the post-initial peak detector is supported in 0/50 seeds. This difference is scientifically important and likely reflects protocol-level differences, especially the open-loop input distribution. We therefore treat C1 as a partial claim-level reproduction: the final deployed gap appears, but the distinctive peak-then-recovery signature is not recovered.

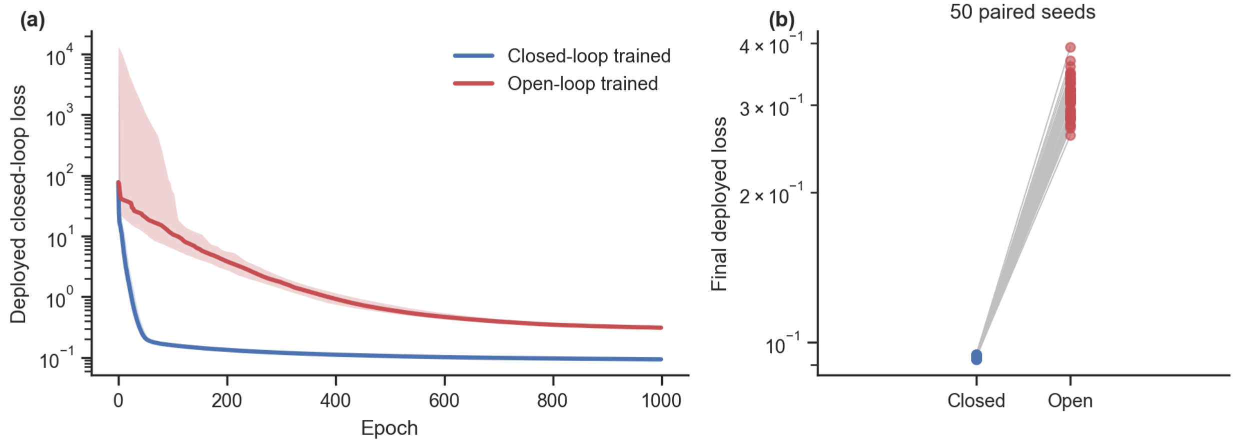

Figure 1: Core C1 reproduction under the original double-integrator setting. (a) Median deployed closed-loop test loss across paired seeds, with interquartile bands. (b) Paired final deployed losses for the same initializations. The final deployed-loss gap appears in 50/50 seeds, but the upstream artifact's post-initial open-loop peak is not recovered by this implementation.

Absolute loss magnitudes are also not directly comparable to the original plots. The upstream artifact's precomputed nonlinear curves end near 6.25–6.34, whereas our final deployed losses are 0.0933 and 0.3119. The discrepancy is consistent with different evaluation conventions, loss normalization, initial-state ranges used in different artifact components, and the independent implementation details listed above. We therefore compare directions, ratios, peak signatures, and seed-level support rather than claiming exact loss-scale reproduction.

6 Result 2: Robustness and Tradeoff Audit

The robustness sweep consists of 195 paired runs: 13 perturbation settings with 15 seeds per setting (Table 3 and Figure 3). The final-loss version of C1 is robust across most perturbations and is supported in 165/195 robustness runs. The two exceptions are informative. With Adam, the final deployed gap almost disappears: mean closed-loop loss is 0.0780 and mean open-loop deployed loss is 0.0798, a relative gap of 2.25 percent, below our 5 percent smallest-effect threshold. This quantifies Ger and Barak's Appendix C.7 prediction that Adam mitigates the structured conflict. With gradient clipping disabled, all 15 `no_clip` runs produce non-finite final losses and are counted as failures rather than negative evidence for the claim.

The stage result should be read narrowly (Figure 2). Our detector is loss-only, whereas the original C2 stages are spectral. A slow-progress phase is visible in 180/195 robustness runs, but the stricter loss-only three-stage criterion is supported in only 1/195. This means a downstream observer of loss curves alone does not recover stable three-stage boundaries; it does not falsify the original spectral stage definition.

The coupled-system stability transition is robust wherever the coupled spectrum is available and the run remains finite. It is supported in 180/195 robustness runs, again failing only in the no-clipping condition. This supports a restricted version of the original mechanistic emphasis in linearizable settings and separates the stability claim from the more fragile stage-boundary claim.

The targeted A1 sweep is more specific than the original union metric suggests (Figure 4). Across six control penalties and 20 seeds per penalty, the union criterion–short-horizon improvement plus either long-horizon loss worsening or increased coupled spectral radius–is supported in 120/120 runs, with mean conditional fraction 0.2814. Splitting the criterion shows that the effect is carried almost entirely by the spectral term: short-horizon improvement plus increased coupled radius is supported in 120/120 runs, whereas the loss-only criterion is supported in 0/120 runs and has mean conditional fraction 0.0038. This distinction prevents the

Table 3: Robustness perturbation grid. The original protocol also has the separate 50-seed core reproduction; the robustness grid repeats the baseline for 15 additional paired seeds so every perturbation has the same seed count.

Perturbation	**Changed value**	**Seeds**
`original`	Baseline double-integrator protocol	50 core; 15 grid
`lr_low`	learning rate 0.003	15
`lr_high`	learning rate 0.03	15
`init_weak`	recurrent initialization scale 0.03	15
`init_strong`	recurrent initialization scale 0.3	15
`feedback_low`	environment feedback strength 0.5	15
`feedback_high`	environment feedback strength 1.5	15
`episode_short`	rollout horizon 25 steps	15
`episode_long`	rollout horizon 100 steps	15
`obs_noise`	observation noise standard deviation 0.05	15
`adam`	optimizer Adam with learning rate 0.001	15
`high_control_penalty`	control penalty 0.05	15
`no_clip`	gradient clipping disabled	15

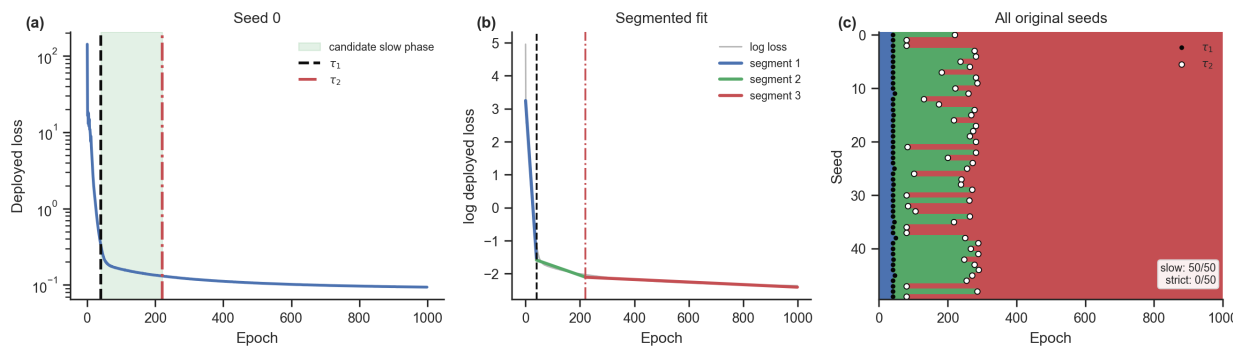

Figure 2: Stage analysis for the original-setting runs. (a,b) A representative seed with candidate boundaries τ_1 and τ_2 and the shaded candidate slow-progress phase. (c) Aggregate stage raster for all 50 original seeds; black dots mark τ_1 and white dots mark τ_2. The segmented detector uses minimum segment length 20 epochs, Stage 1 slope < -0.02, Stage 2 slope magnitude below $\max(0.35|\text{slope}_1|, 0.004)$, and Stage 3 reacceleration slope $< \text{slope}_2 - 0.002$. It finds rapid-improvement plus slow-progress evidence in 50/50 seeds but strict three-stage support in 0/50.

audit from overstating a behavioral loss tradeoff: the detected A1 signal is a stability-cost signal, not a deployed-loss reversal.

7 Result 3: Which Diagnostics are Predictive?

The original-setting coupled-system stability transition is detected in 50/50 seeds (Figure 5), but its timing does not support the strongest alignment reading of C3. The final coupled spectral radius is finite and below the stability threshold in every original run, and the median crossing epoch is 54. This aligns with the first loss-only boundary, not with the later candidate plateau exit: segmented loss changepoints have median boundaries 40 and 255, so the stability crossing precedes τ_2 by a median of 201 epochs. Thus our data support the restricted statement that coupled spectra detect an early stability transition more directly than RNN-only spectra, but they do not reproduce the original visual claim that the end of Stage 2, as seen in loss, coincides with the stability phase transition.

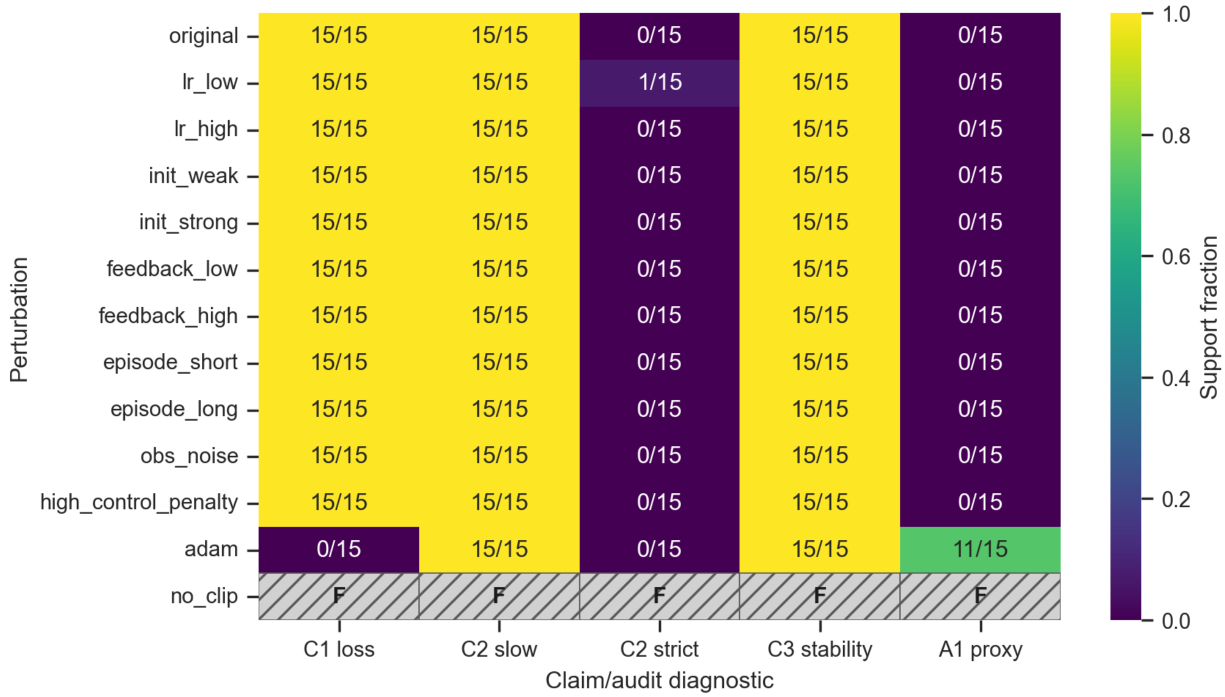

Figure 3: Robustness heatmap over perturbation settings and claim diagnostics. C1 and C3 are stable across most finite settings, C2 is robust only under the broad slow-progress criterion, the weak A1 proxy is not decisive outside the targeted multi-horizon sweep, and disabling gradient clipping produces non-finite failures marked as F rather than zero support.

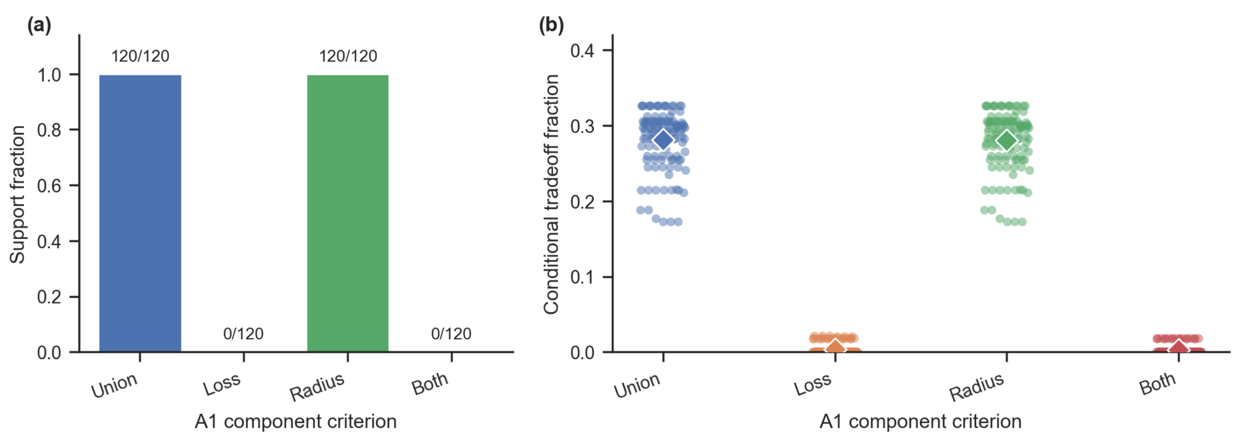

Figure 4: Targeted A1 component analysis over 120 paired tradeoff runs. (a) Support counts for the union criterion, loss-only criterion, radius criterion, and the intersection of loss and radius criteria. (b) Seed-level conditional fractions among short-horizon improvement intervals. The broad union result is 120/120, but the loss-only short-vs-long criterion is 0/120; the signal is dominated by coupled-radius increases. Diamonds show means.

The spectral diagnostic is architecture-dependent in the current tooling, not in principle. For linear, tanh, and low-rank controllers on linear tasks, the coupled transition matrix can be constructed explicitly. We do not report coupled spectral diagnostics for GRU or ring variants because this implementation does not yet

Table 4: C2 threshold sensitivity on the 50 original-setting seeds. Slow-progress support is stable, but strict three-stage support remains absent across looser, stricter, and shorter-segment changepoint variants.

Detector variant	Slow phase	Strict three-stage	Median τ_1, τ_2
Main segmented	50/50	0/50	40, 255
Loose segmented	50/50	0/50	40, 255
Strict segmented	47/50	0/50	40, 255
Short-segment binary style	49/50	0/50	20, 70
Very loose reacceleration	50/50	0/50	40, 255

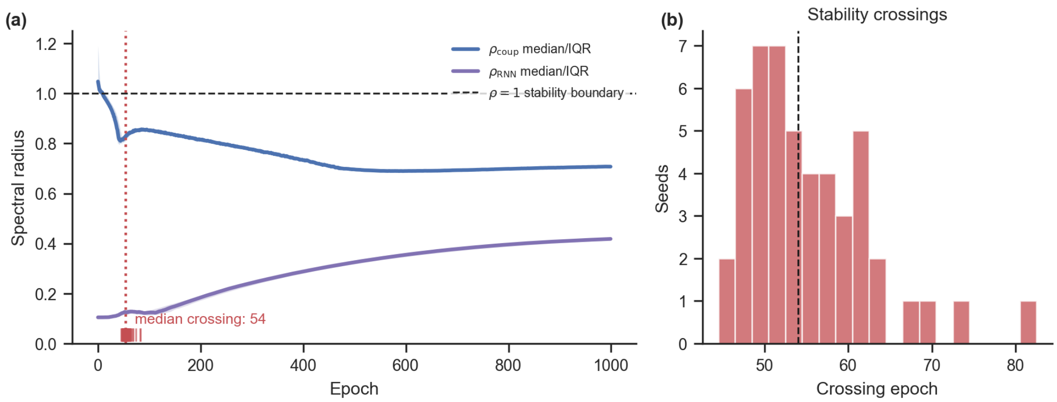

Figure 5: Coupled-system spectral analysis for the original setting. (a) Median coupled and RNN-only spectral radii across 50 seeds, with interquartile bands; the dashed line marks the $\rho = 1$ stability boundary and tick marks show per-seed crossing epochs. (b) Distribution of stability-crossing epochs. The coupled spectral radius crosses below the stability boundary early in training, whereas the RNN-only spectral radius remains far from the relevant stability threshold.

compute local Jacobian spectra for gated nonlinear controllers. This is a limitation of the audit code: local linearization should be added before making claims about C3 transfer to GRUs or nonlinear task families.

The most important diagnostic distinction concerns C2. The coupled stability crossing is robust, but our loss-only detector is not the original spectral stage detector. In the original setting, a slow-progress phase is present in 50/50 seeds, while segmented regression finds median changepoint boundaries at epochs 40 and 255 with slopes -0.1131, -0.00229, and -0.00036. The third segment does not show post-plateau reacceleration under any threshold variant in Table 4. This result should be interpreted as a negative result for loss-only stage recovery, and as motivation to log full coupled eigenvalue trajectories, σ_{zm}, and λ_3 in future spectral reproductions.

8 Result 4: Does the Phenomenon Generalize?

Generalization is hierarchical rather than uniform (Figure 6). Across 90 completed architecture/task extension runs, C1 deployed-loss divergence is supported in 42/90 runs. Architecture transfer within the double-integrator-style control family is strong: tanh RNN, GRU, and low-rank variants each show C1 support in 10/10 seeds. Task transfer to tracking is partial, with deployed-loss divergence in 8/10 seeds. Task-family transfer to the harder partial-observation moving-target ring variants is weak, with 3/20 GRU seeds and 1/20 tanh seeds passing the deployed-loss criterion.

The simple ring variant is best viewed as a sanity check, not as a hard generalization test: it gives the learner sin(error), cos(error), and velocity with a fixed target, making the mapping nearly memoryless. As expected, it has 0/10 supporting seeds and mean deployed-loss gap 5.4×10^{-4}. The harder ring follow-up is fairer but still weak, with mean deployed-loss gaps 0.0837 for GRU and 0.0085 for tanh. These results sharpen A2: the

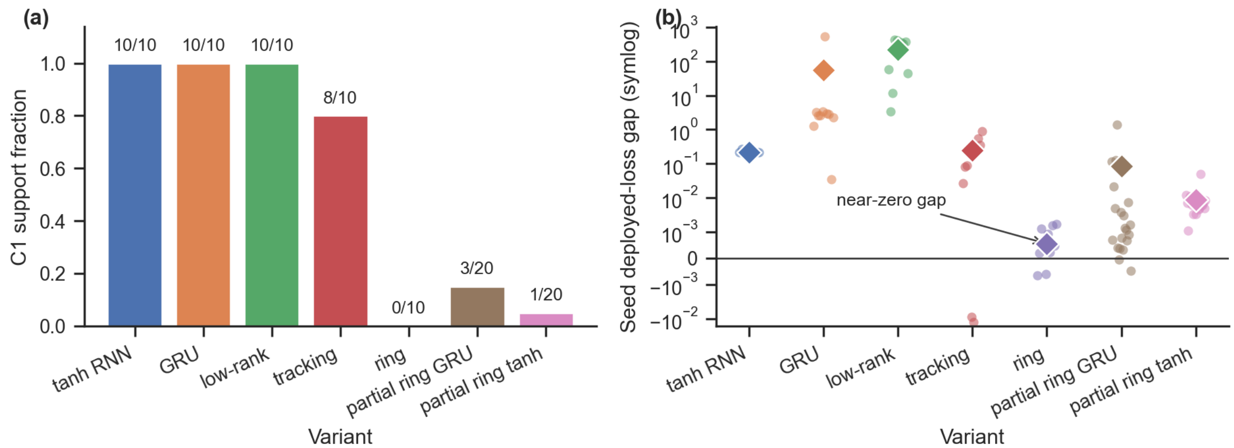

Figure 6: Generalization results across architecture and task variants. Standard variants use $n = 10$ paired seeds; the harder partial-observation ring variants use $n = 20$ paired seeds. (a) C1 support fractions with exact seed counts. (b) Seed-level deployed-loss gaps with mean diamonds on a symmetric-log scale, which permits tiny or negative gaps while still showing large architecture gaps. The simple ring is a sanity check for a nearly memoryless setting; the partial-observation ring variants are the stronger path-integration tests.

closed/open divergence transfers across architecture changes on the double-integrator-style control problem, but it does not automatically transfer to path-integration variants.

C3 diagnostic transfer is only tested where the coupled spectrum is implemented. It is supported in 10/10 tanh RNN runs and 10/10 low-rank runs, but not reported for GRU or ring runs because the current spectral code does not linearize gated nonlinear controllers. Local Jacobian spectra for gated nonlinear controllers are needed before claiming diagnostic transfer to GRUs. This is another reporting lesson: a generalization claim about stability mechanisms should distinguish performance transfer from diagnostic transfer.

9 Claim-Level Audit Matrix

Table 5 is the main summary of the paper. It separates original claims C1–C3 from audit questions A1–A2, and attaches each to evidence, robustness or generalization results, and an actionable lesson.

10 Actionable Lessons for Closed-Loop RNN Studies

The goal of the audit is to produce reusable reporting practices. We propose the following checklist for closed-loop RNN papers.

Report deployed closed-loop performance and peak-over-training signatures. The central risk is that a model can imitate a teacher under fixed inputs but fail when its own actions determine future inputs; final loss alone can miss transient peaks that define the qualitative phenomenon.

Use paired-seed comparisons when contrasting open-loop and closed-loop training. The clean comparison uses the same initialization, architecture, optimizer budget, and evaluation rollouts while changing only the training regime.

Define learning stages in the diagnostic space used by the claim. Spectral stage claims should report eigenvalue criteria, order parameters, and boundary epochs, while loss-only changepoints should be labeled as downstream observables.

Table 5: Claim-level audit matrix for completed runs. C1–C3 are original claims under reproduction. A1–A2 are audit questions introduced here to test robustness, short/long horizon tradeoffs, and broader generalization.

Item	**Evidence type**	**Original protocol**	**A1 audit**	**A2 audit**	**Lesson**
C1: open/closed divergence	Final deployed loss and peak signature	Partial: final gap 50/50, post-initial peak 0/50	Robust in most perturbations; Adam mitigation matches original Appendix C.7; no clipping fails	Strong architecture transfer, tracking partial, path-integration weak	Report deployed loss and peak-overtraining signatures.
C2: stage structure	Loss-only detector vs spectral definition	Loss-only slow phase 50/50; spectral C2 not tested	Loss-only strict exit boundary fragile	Loss-only stages vary by task	Stage claims need diagnostics in the same space as the claim.
C3: coupled stability	Coupled spectral radius and timing	Crossing 50/50; crossing precedes loss τ_2 by median 201 epochs	Robust finite crossings; alignment remains unresolved	Holds for tanh/low-rank; local Jacobians not implemented for GRU/ring	Report coupled spectrum and its timing relative to stage boundaries.
A1: protocol robustness and tradeoff	Perturbation grid and multi-horizon component metric	N/A	Partial: union/radius 120/120, loss-only 0/120	N/A	Robustness claims need stress tests, explicit horizons, and component criteria.
A2: broader generalization	Architecture and task extensions	N/A	N/A	Partial: architecture transfer strong, tracking partial, ring weak	Closed-loop effects depend on task structure, environment coupling, and observability.

Analyze the coupled agent-environment system. RNN-only spectra can miss stability changes introduced by the environment. Coupled diagnostics should be reported whenever a mechanistic stability claim is made.

Report feedback strength and rollout horizon. These parameters determine how strongly actions affect future observations and can control whether the closed-loop phenomenon appears.

Report failure rates, not just average curves. Means can hide unstable runs, recovery failures, or seed-dependent plateaus.

Include at least one stress test. Optimizer, initialization, noise, feedback strength, and horizon perturbations are inexpensive compared with the interpretive cost of an untested mechanistic claim.

11 Discussion

This project treats Ger and Barak's framework as a case study in reproducibility for interactive recurrent systems. The completed audit supports a final deployed-loss divergence under our independent original-setting implementation and detects coupled stability crossings in all original-setting seeds. It also identifies important limits. First, the original spectral stage claim is not tested by our loss-only detector; the detector shows that loss curves alone do not recover a robust second boundary. Second, the coupled stability crossing occurs early relative to the later loss changepoint, so our data do not reproduce the strongest visual alignment between Stage 2 end and stability transition. Third, A1 shows a stability-cost signal through coupled-radius increases, but not a loss-only short-vs-long horizon reversal. Fourth, generalization is hierarchical: architecture transfer within the double-integrator-style task is strong, tracking transfer is partial, and path-integration transfer is weak.

These mixed results are the reason a claim-level audit is useful. A figure-level reproduction might emphasize only that a closed/open loss gap appears. A methodological audit can say more: the upstream open-loop peak is not reproduced by our independent open-loop protocol, Adam reduces the gap as predicted by the original appendix, missing gradient clipping can produce outright failure, path-integration-style variants can erase the gap, spectra for gated controllers require local Jacobian tooling, and stage claims should be accompanied by spectral boundaries rather than visual annotations alone. This is especially important for computational neuroscience, where RNNs are often used as mechanistic models and where closed-loop behavior is closer to biological learning than fixed-dataset supervision.

12 Limitations

The audit does not cover every RNN architecture, every control task, or full reinforcement-learning algorithms. Training uses direct gradients through the rollout, not sparse-reward or likelihood-ratio RL. The path-integration extensions are intentionally lightweight, and the simple ring variant is a sanity check for a nearly memoryless setting. The A1 tradeoff metric is interval-level and diagnostic: it shows that short-horizon improvement often coincides with increased coupled spectral radius, but the loss-only horizon tradeoff is not supported. Finally, coupled spectral analysis is implemented for explicitly constructed coupled matrices, but not yet for local Jacobians of GRUs or other gated nonlinear controllers.

13 Artifact

The accompanying anonymous repository provides a config-driven implementation with smoke and full-run modes, raw CSV outputs, claim tables, and figure-generation scripts. The artifact is organized around the audit protocol rather than notebook-only reproduction: each experiment records the config, seed, device, metrics, and per-epoch time series. During double-blind review, the supplementary archive omits public repository URLs, Software Heritage origin links, Git remotes, commit history, local logs, and machine-specific paths; a public archival link should be added only after de-anonymization. Table 6 lists the main entry points.

Table 6: Artifact entry points. Full runtimes are approximate for one NVIDIA L40 GPU and are intended for planning rather than benchmarking.

Component	File/script	Produces	Runtime	Expected output
Smoke validation	`scripts/run_smoke.sh`	Tiny raw outputs, tables, figures	$<$ 10 min	`results/raw/smoke_*`
Full audit	`scripts/run_full_audit.sh`	Core, robustness, generalization, figures	16–18 h	sweep summaries, figures
Targeted C2/A1/A2	`scripts/run_targeted_c2_a1_a2.sh`	Tradeoff, hard ring, changepoints	8–10 h	targeted summaries
Claim tables	`make_claim_tables` module	Claim matrix	$<$ 1 min	`claim_matrix.csv`
Supplemental tables	`supplemental_audit_tables` module	Run accounting, C2 sensitivity, A1 split, peak/alignment checks	$<$ 1 min	processed CSVs
Figures	`make_all_figures` module	Paper figures	$<$ 1 min	`figure_*.png`

References

Yoav Ger and Omri Barak. Learning dynamics of RNNs in closed-loop environments. In *Advances in Neural Information Processing Systems*, poster, 2025. OpenReview: `https://openreview.net/forum?id=P63bBMXCIH`.

Omri Barak. Recurrent neural networks as versatile tools of neuroscience research. *Current Opinion in Neurobiology*, 46:1–6, 2017.

Samy Bengio, Oriol Vinyals, Navdeep Jaitly, and Noam Shazeer. Scheduled sampling for sequence prediction with recurrent neural networks. In *Advances in Neural Information Processing Systems*, 2015.

Peter Henderson, Riashat Islam, Philip Bachman, Joelle Pineau, Doina Precup, and David Meger. Deep reinforcement learning that matters. In *Proceedings of the AAAI Conference on Artificial Intelligence*, 2018.

Niru Maheswaranathan, Alex H. Williams, Matthew D. Golub, Surya Ganguli, and David Sussillo. Universality and individuality in neural dynamics across large populations of recurrent networks. In *Advances in Neural Information Processing Systems*, 2019.

Valerio Mante, David Sussillo, Krishna V. Shenoy, and William T. Newsome. Context-dependent computation by recurrent dynamics in prefrontal cortex. *Nature*, 503:78–84, 2013.

Joelle Pineau, Philippe Vincent-Lamarre, Koustuv Sinha, Vincent Lariviere, Alina Beygelzimer, Florence d'Alché-Buc, Emily Fox, and Hugo Larochelle. Improving reproducibility in machine learning research: a report from the NeurIPS 2019 reproducibility program. *Journal of Machine Learning Research*, 22(164):1–20, 2021.

Stéphane Ross, Geoffrey Gordon, and Drew Bagnell. A reduction of imitation learning and structured prediction to no-regret online learning. In *Proceedings of the Fourteenth International Conference on Artificial Intelligence and Statistics*, pages 627–635, 2011.

David Sussillo and Omri Barak. Opening the black box: low-dimensional dynamics in high-dimensional recurrent neural networks. *Neural Computation*, 25(3):626–649, 2013.

Ian Goodfellow, Yoshua Bengio, and Aaron Courville. *Deep Learning*. MIT Press, 2016.

A Metric Definitions

Let s_t be the environment state, $o_t = Cs_t + \epsilon_t$ the observation, h_t the recurrent state, and $u_t = f_\theta(o_t, h_t)$ the action. Closed-loop deployment follows $s_{t+1} = F(s_t, u_t)$ and $h_{t+1} = G_\theta(o_t, h_t)$. The deployed loss is

$$L_{\text{deploy}}(\theta) = \frac{1}{T}\sum_{t=0}^{T-1}\left[\ell_s(s_t, s_t^\star) + \lambda_u \|u_t\|_2^2\right], \tag{1}$$

while teacher-forced imitation loss compares the learner action to the teacher action on teacher-generated observations:

$$L_{\text{TF}}(\theta;\theta^\star) = \frac{1}{T}\sum_{t=0}^{T-1}\|f_\theta(o_t^\star, h_t) - f_{\theta^\star}(o_t^\star, h_t^\star)\|_2^2. \tag{2}$$

For linearizable double-integrator settings we estimate an effective feedback gain by ridge regression, $\widehat{K}_t = (X^\top X + \alpha I)^{-1}X^\top U$, and report the closed/open gain distance $D_K = (\sum_{t\in\mathcal{E}}\|\widehat{K}_t^{\text{CL}} - \widehat{K}_t^{\text{OL}}\|_F^2)^{1/2}$.

For linear RNN controllers and linear environments, with $s_{t+1} = As_t + Bu_t$, $u_t = W_{ho}h_t$, and $h_{t+1} = (1-\eta)h_t + \eta(W_{hh}h_t + W_{ih}Cs_{t+1})$, the coupled transition matrix over $z_t = [s_t^\top, h_t^\top]^\top$ is

$$P_\theta = \begin{bmatrix} A & BW_{ho} \\ \eta W_{ih}CA & (1-\eta)I + \eta W_{hh} + \eta W_{ih}CBW_{ho}\end{bmatrix}. \tag{3}$$

We report $\rho_{\text{RNN}} = \max_i |\lambda_i(W_{hh})|$ and $\rho_{\text{coup}} = \max_i |\lambda_i(P_\theta)|$. The stability-crossing epoch is the first epoch at which $\rho_{\text{coup}} < 1$ for a minimum persistence window.

The open-loop peak signature is evaluated over the deployed-loss trajectory across training epochs. We count a post-initial peak when $\arg\max_e L_{\text{deploy},e}^{\text{OL}} > 0$ and $\max_e L_{\text{deploy},e}^{\text{OL}} > 1.5L_{\text{deploy},0}^{\text{OL}}$; we also report peak/initial and peak/final ratios.

Stage labels use smoothed log deployed loss $y_t = \log(\max(L_{\text{deploy},t}, \varepsilon))$. This is explicitly a loss-only downstream detector, not the original spectral definition. It looks for a fast initial negative slope, a sustained slow-progress interval, and a later reacceleration; the segmented variant fits three piecewise-linear segments and requires Stage 1 slope < -0.02, Stage 2 slope magnitude below $\max(0.35|\text{slope}_1|, 0.004)$, and Stage 3 slope $< \text{slope}_2 - 0.002$.

For A1, define a myopic-improvement interval by $\Delta \log L_e^{(T_s)} < -\delta_{\text{short}}$. We report four component criteria: union $I_s \wedge (I_\ell \vee I_\rho)$, loss $I_s \wedge I_\ell$, radius $I_s \wedge I_\rho$, and both $I_s \wedge I_\ell \wedge I_\rho$, where I_ℓ is $\Delta \log L_e^{(T_\ell)} > \delta_{\text{long}}$ and I_ρ is $\Delta\rho_{\text{coup},e} > 0$. In the targeted sweep, $T_s = 10$, $T_\ell = 200$, and $\delta_{\text{short}} = \delta_{\text{long}} = 0.005$.

For scalar paired metric M, the open/closed effect is $\Delta_i(M) = M_i^{\text{OL}} - M_i^{\text{CL}}$. We report mean, nonparametric bootstrap 95 percent CI, seed-level SD/IQR where useful, direction agreement $N^{-1}\sum_i \mathbf{1}[\Delta_i(M) > 0]$, and binary claim-support rate $S(c) = N^{-1}\sum_i q_i(c)$.